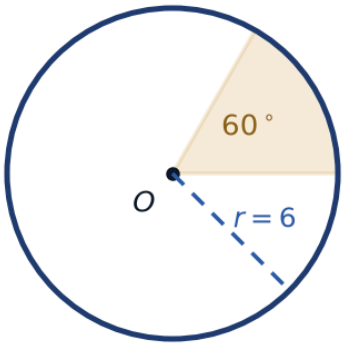
Functions of Several Variables & Partial Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

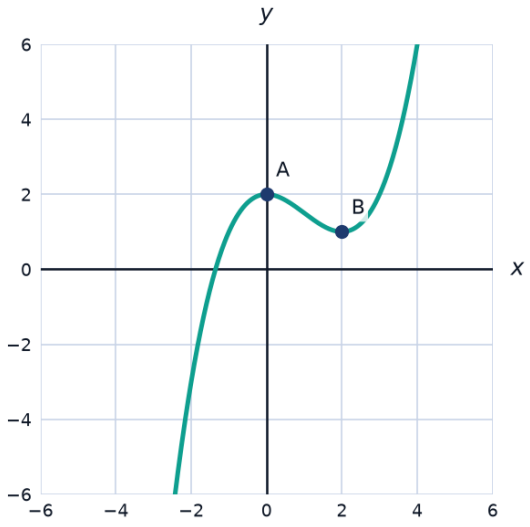
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Verify Clairaut's Theorem on equality of mixed partial derivatives for the function $f(x, y) = \cos(2x - 3y) + x^2y^3$ by computing f_{xy} and f_{yx} directly.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $f(x, y) = x^3e^{2y} - y\ln(x)$ defined on $x > 0$. Compute the first-order partial derivatives $f_x(x, y)$ and $f_y(x, y)$, and evaluate $f_x(1, 0)$ and $f_y(1, 0)$.



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3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A temperature distribution is given by $T(x, y, z) = \frac{120}{1 + x^2 + 2y^2 + 3z^2}$. At the point $P(1, 1, 1)$, find: (a) the direction of steepest temperature increase, (b) the maximum rate of increase, and (c) the equation of the tangent plane to the isothermal level surface passing through P.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

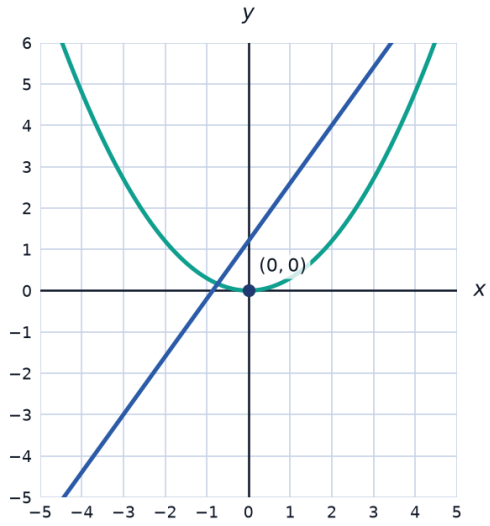
Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve C is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region D enclosed by C, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

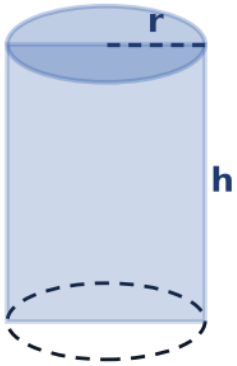
Let $z = f(x, y)$ be a differentiable function, with polar coordinates transformation $x = r\cos\theta$ and $y = r\sin\theta$. Express $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, and prove that: $\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = ||ablaz||^2$.



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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

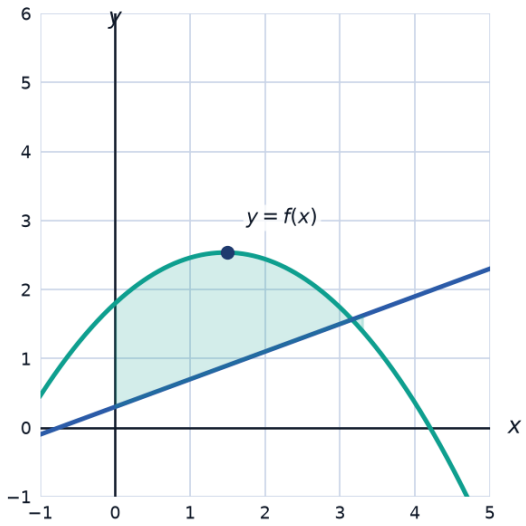
Use the method of Lagrange multipliers to find the maximum volume of a rectangular box with faces parallel to the coordinate planes inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

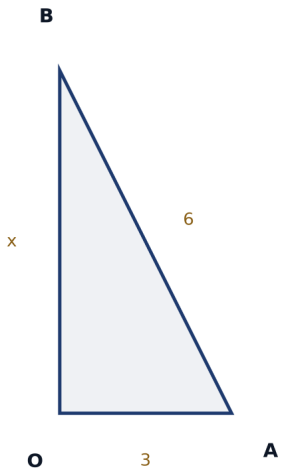
Derive the Gaussian integral $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$ from first principles by evaluating the double integral $I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dA$ in polar coordinates.



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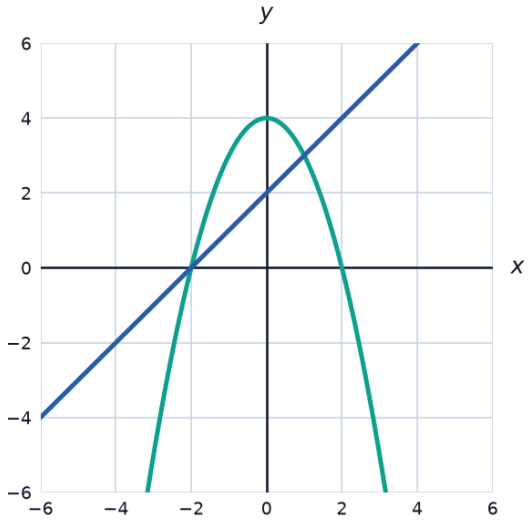
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\,dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta}\int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\cdot r\,dr\,d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2}\int_0^{\frac{6}{\sin\theta+2\cos\theta}} (r\cos\theta + r\sin\theta)\cdot r\,dr\,d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the non-elementary integral $\int_0^1\int_y^1 e^{x^2}dx\,dy$ by reversing the order of integration.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use Stokes' Theorem to evaluate $\oint_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y^3, x^3, -z^3\rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1$, oriented counterclockwise when viewed from above. Verify surface independence by comparing the flux of $abla \times \mathbf{F}$ through the flat disk $S_1 : x^2 + y^2 \le 1, z = 1$ and the paraboloid cap $S_2 : z = 2 - (x^2 + y^2)$.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A particle's position in three-dimensional space is given by the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where t represents time. The velocity of the particle is given by $\mathbf{v}(t) = \mathbf{r}'(t)$. Suppose the velocity components are related by the partial differential equations: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. If at time $t = 0$, the particle is at the origin $(0, 0, 0)$ and its velocity is $\mathbf{v}(0) = \langle 5, -2, 0 \rangle$, determine the value of $\frac{\partial^2 v_x}{\partial t^2}$ at $t = 0$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + y\sin z,\ y^3 + z\cos x,\ z^3 + xy \rangle$ across the closed boundary surface of the solid cylinder $E : x^2 + y^2 \le a^2,\ 0 \le z \le h$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use the Green's Theorem line integral formula for area, $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, to derive the area enclosed by the hypocycloid (astroid) $\mathbf{r}(t) = \langle a\cos^3 t,\ a\sin^3 t \rangle$ for $0 \le t \le 2\pi$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Double Integrals

Let the function $f(x, y)$ be defined as $f(x, y) = 2x^2 - 3y^2$. Calculate the partial derivative of f with respect to y at the point $(2, 3)$.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let S be the closed surface of the unit cube defined by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through S is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume V enclosed by S. Calculate this flux by evaluating $\iiint_V abla \cdot \mathbf{F}\, dV$.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use Green's Theorem to evaluate $\oint_C (y^2 - 7y)\,dx + (2xy + 2x)\,dy$, where C is the positively oriented boundary of the region D enclosed between the parabolas $y = x^2$ and $x = y^2$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Double Integrals

Use Green's Theorem to evaluate the circulation $\oint_C (3y - e^{\sin x})\,dx + (7x + \sqrt{y^4 + 1})\,dy$, where C is the circle $x^2 + y^2 = 9$ oriented counterclockwise.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the function $f(x, y) = 3x^2 - 5y^3$. Find the partial derivative of f with respect to y evaluated at the point $(2, -1)$, i.e., $f_y(2, -1)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle x, y, z \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the domain of the function $f(x, y) = \ln(9 - x^2 - y^2)$.

a. The interior of the circle $x^2 + y^2 < 9$

b. The exterior of the circle $x^2 + y^2 > 9$

c. The closed disk $x^2 + y^2 \leq 9$

d. All of $\mathbb{R}^2$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What geometric curves are the level curves of $z = x^2 + y^2$ for $c > 0$?

a. Concentric circles centered at $(0, 0)$

b. Lines of slope 1

c. Parabolas opening upward

d. Hyperbolas

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the range of $g(x, y) = \sqrt{16 - x^2 - y^2}$?

a. $[0, 4]$

b. $[0, 16]$

c. $[-4, 4]$

d. $[0, \infty)$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the partial derivative $f_x(x, y)$ for $f(x, y) = 3x^3y^2 - 5x^2 + 4y^3$.

a. $9x^2y^2 - 10x$

b. $6x^3y + 12y^2$

c. $9x^2y^2 - 10x + 12y^2$

d. $3x^2y^2 - 10x$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does Clairaut's Theorem state about mixed second-order partial derivatives f_{xy} and f_{yx}?

a. If both f_{xy} and f_{yx} are continuous on an open region, then $f_{xy} = f_{yx}$

b. Mixed partials are always equal to 0

c. The sum $f_{xy} + f_{yx} = 0$

d. Equality holds only for polynomials

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For $f(x, y) = x^3y^4$, verify Clairaut's Theorem by finding f_{xy}.

a. $12x^2y^3$

b. $3x^2y^4$

c. $4x^3y^3$

d. $6xy^2$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the equation of the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0)?

a. $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

b. $z - z_0 = f_x(x - x_0) - f_y(y - y_0)$

c. $z = f_x x_0 + f_y y_0$

d. $(x - x_0) + (y - y_0) + (z - z_0) = 0$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the equation of the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.

a. $z - 5 = 2(x - 1) + 4(y - 2) \Leftrightarrow 2x + 4y - z = 5$

b. $2x + 4y + z = 15$

c. $x + 2y - z = 0$

d. $4x + 2y - z = 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is a normal vector to the surface $z = f(x, y)$ at (x_0, y_0)?

a. $\mathbf{n} = [f_x(x_0, y_0), f_y(x_0, y_0), -1]$

b. $\mathbf{n} = [f_x, f_y, 1]$

c. $\mathbf{n} = [1, 1, 1]$

d. $\mathbf{n} = [-f_y, f_x, 0]$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x, y) = 5x^3 - 2y^2 + 7$. Compute the value of the partial derivative $f_y(2, -3)$.

a. 20

b. -12

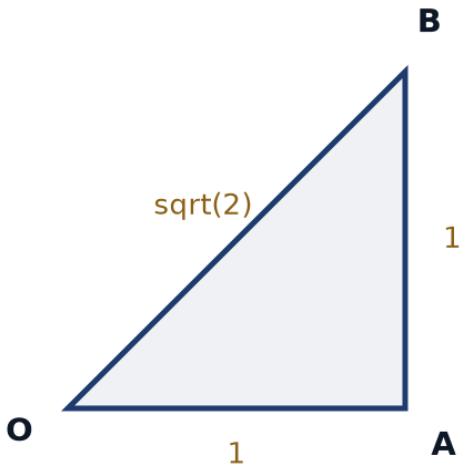
c. 12

d. -20

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the double integral $\iint_R xy\,dA$ where R is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.



a. 1/12

b. 1/8

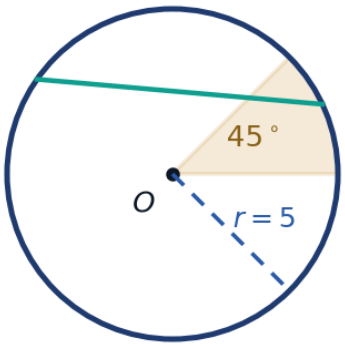
c. 1/6

d. 1/4

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32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Consider the region R in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\, dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\, dr\, d\theta = \int_0^{\pi/4} A\, d\theta$, find the value of A.



a. 4

b. $\pi/4$

c. 16

d. 2

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For $f(x, y) = 4x^2y^2 + 3xy$, find the mixed partial $f_{xy}(1, -2)$.

a. -32

b. 26

c. 29

d. -29

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.

a. 10

b. 13

c. 26

d. 52

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the region R in the xy-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\,dA$.

a. $\frac{1}{12}$

b. $\frac{1}{10}$

c. $\frac{1}{8}$

d. $\frac{1}{6}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

a. 6

b. 12

c. -6

d. 15

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Under the change of variables $x = 4u - 2v$, $y = -u + 4v$, a region S in the uv-plane with area 3 maps to a region R. Find the area of R.

a. 3

b. 14

c. 42

d. 24

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Double Integrals

Euler's theorem for homogeneous functions: if $f(tx, ty) = t^n f(x, y)$, then $xf_x + yf_y =$

a. f

b. nf

c. $n(n-1)f$

d. 0

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Gradient descent step: $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf(\mathbf{x}_n)$. For $f = x^2 + y^2$ at $(1, 1)$, $\alpha = 0.5$:

a. $(0.5, 0.5)$

b. $(0, 0)$

c. $(-1, -1)$

d. $(1.5, 1.5)$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, y \rangle$. Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise. Use Green's Theorem.

a. 0

b. 8π

c. 4π

d. 16π

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Double Integrals

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.

a. (2, 3)

b. (-2, -3)

c. (4, 6)

d. (0, 0)

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the solid region E in the first octant bounded by the coordinate planes and the surface $z = 4 - x^2 - y^2$. The volume of this region is given by the triple integral $V = \iiint_E dV$. We are given that the volume can also be expressed in cylindrical coordinates as $V = \int_0^{\pi/2} \int_0^{\sqrt{a}} \int_0^{4 - r^2} r \, dz \, dr \, d\theta$. Find the value of the parameter a.

a. 2

b. 4

c. 16

d. 8

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the solid region E bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of E is given by the triple integral $V = \iiint_E dV$. If E is expressed in spherical coordinates, the integral becomes $\int_\alpha^\beta \int_\gamma^\delta \int_{\rho_1}^{\rho_2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. Determine the value of α, the lower limit of integration for θ.

a. 0

b. $\frac{\pi}{2}$

c. π

d. 2π

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Double Integrals

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

a. 1

b. 2

c. 3

d. 4

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the solid region E defined by $0 \le x \le 1$, $0 \le y \le 2$, and $0 \le z \le 3$. Evaluate the triple integral of the function $f(x, y, z) = 2x + 3y$ over the region E.

a. 18

b. 15

c. 21

d. 24

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